

Akhil Bejjanki

Pleasanton, California | 510-807-1095 | akhilbejjanki30@gmail.com | U.S. Citizen

Education

Georgia Institute of Technology | Atlanta, GA

Graduation: Dec 2027

Bachelor of Science in Computer Engineering, GPA 4.0/4.0

Skills

Programming: Java, Python, C++, HTML, CSS, SQL, Git, Swift, CI/CD

Embedded/Hardware Development: RaspberryPi, Arduino, AutoDesk Fusion, Onshape, Rust, RTIC, Altium

Frameworks/Databases: React, MongoDB, Typescript, Node.js, Spring, FastAPI, Docker, GCP, Vercel

Experience

Mechatronics & Motivation @ Flavin Neuromachines Lab | Atlanta, Georgia

December 2025 – Present

Undergraduate Researcher

- Designed flexible PCBs in Altium for wearable haptic ring and EMG device, enabling compact biosignal acquisition and feedback.
- Developed embedded firmware for EMG sensors with Bluetooth IoT connectivity, performing real-time muscle signal processing via BLE and streaming data to a custom visualization dashboard using Swift.
- Implemented real-time biofeedback system integrating EMG sensing and adaptive haptic feedback; conducted user experiments with K-12 students demonstrating improved engagement through survey-based evaluation.

RoboCup | Atlanta, Georgia

August 2025 – Present

Firmware Engineer

- Developed embedded firmware in Rust for autonomous robotic systems, implementing motor control, sensor integration, and real-time hardware communication via I2C and ADC interfaces
- Programmed an RTIC-based cooling system in Rust integrating fail-safe logic to prevent overheating and ensure system reliability
- Wrote firmware for multi-agent coordination systems, enabling real-time communication across autonomous robots.

Hydra | Pleasanton, California

May 2025 – August 2025

Software Development Intern

- Collaborated with engineers to develop a web platform connecting open-source contributors with companies, implementing responsive UI components and improving feature usability across the application.
- Contributed to platform architecture and researched AI-driven product strategies for scalable digital business models

Ikey studios | Remote

August 2023 – July 2024

Software Intern

- Deployed production website serving 100+ monthly users, enhancing navigation, usability through front-end development.
- Implemented multilingual functionality and built React-based UI components for an online shop, enhancing accessibility and supporting a more scalable front-end architecture

Projects

Smart Path AI | Big Data Big Impact @ Georgia Tech

August 2025 – Present

AI + Image Processing Lead

- Architected GraphRAG pipeline using LangChain, OpenAI, and Neo4j to transform PDFs into interactive knowledge graphs and dynamic quizzes that adjust question difficulty and concept sequencing to optimize learning.
- Implemented handwritten PDF processing by integrating Google Cloud Document AI OCR into a FastAPI-based AI pipeline, enabling extraction of handwritten notes and feeding parsed text into LLM and improved document processing speed by 50%..
- Engineered an adaptive learning system through a MongoDB-backed user model, leveraging stored confidence scores and spaced repetition algorithms to generate personalized quizzes in real time based on student performance.

Material Sorter

April 2024 – May 2024

Engineer and Programmer

- Designed and programmed an automated device that sorts recycled-material marbles into bins with 100% accuracy
- Modeled and 3D-printed custom input mechanism using AutoCAD, streamlining sorting and ensuring single-marble precision
- Delivered a professional presentation and live demonstration of device to CEOs, showcasing design and real-world functionality

Leadership

AVHS Engineers | Programming Captain

August 2022 – May 2025

- Led team to states by designing, building, and programming three unique competition robots featuring pneumatics, catapults, and custom drivetrains
- Guided programming in C++ for autonomous and driver-control systems while mentoring new members in coding through developing hands-on tutorials for robotics fundamentals.